

AI, Human Cognition and Knowledge Collapse

Acemoglu, Kong & Ozdaglar (2026)

William Gradost

Artificial Intelligence and Economic Modeling · Repository 4

<https://github.com/wbgradost/ai-04-acemoglu>

1 · The paper and the agent's problem

Question. Can accurate agentic AI improve today's decisions while eroding the human learning that sustains tomorrow's collective knowledge?

A short-lived, atomistic agent chooses $e \geq 0$, taking public precision X_t and agentic precision τ_A as given. Effort produces both context-specific and “thin” public information, but the latter is an externality.

$$Y = \sigma^{-2} + \lambda_I e + \tau_A \quad (4)$$

Under Assumption 1 ($\Delta_I = 0$, $\Delta_X > 0$),

$$\max_{e \geq 0} f(0, 0) + G(X_t)\Delta_G + G(X_t)G(Y)\Delta_X - \frac{\varepsilon}{\varepsilon + 1} e^{(\varepsilon+1)/\varepsilon}. \quad (6)$$

$G(\tau) = 2\Phi(\sqrt{\tau}) - 1$: probability that a Gaussian prediction error with precision τ lies within the success band.

Observation 1 (pp. 14–15). For positive precisions, $\lambda_I > 0$, and Assumption 1,

$$U_e = \Delta_X G(X_t) \lambda_I g(Y) - e^{1/\varepsilon},$$

$$U_{eX} = \Delta_X \lambda_I g(X_t) g(Y) > 0, \quad U_{e\tau_A} = \Delta_X G(X_t) \lambda_I g'(Y) < 0,$$

since $g = G' > 0$ and $g'(\tau) = -\frac{1}{2}(1 + \tau^{-1})g(\tau) < 0$.

Mechanism. General knowledge makes context-specific learning useful; AI supplies context-specific precision directly, where returns are diminishing.

Dynamic condition. With $\varepsilon > 4$, zero knowledge is locally stable; if $\tau_A > \tau_A^c$, it is uniquely and globally stable (Proposition 5). If $\varepsilon < 4$, every $X_1 > 0$ converges instead to the high-knowledge state (Proposition 3).

3 · What I did

- ▶ Reconstructed equations (3)–(6) and the effort FOC from the May 5, 2026 MIT copy (69 pages), keeping equation and page references.
- ▶ Derived both cross-partial from $g(\tau) = (2\pi)^{-1/2}e^{-\tau/2}\tau^{-1/2}$, including the corner $X_t = 0 \Rightarrow e = 0$.
- ▶ Traced the feedback loop:

$$\tau_A \uparrow \Rightarrow e_t \downarrow \Rightarrow X_{t+1} \downarrow \Rightarrow e_{t+1} \downarrow.$$

- ▶ Audited welfare: under Assumption 2, $\sigma^{-2} \geq \sqrt{2} - 1$, Propositions 10–11 imply a finite, single-peaked welfare optimum in AI precision.
- ▶ Checked Section 5 and the appendix before proposing an extension: aggregation, synthetic data, and effort separability are already covered; the production restriction $\Delta_I = 0$ is not relaxed.

First-pass intuition to audit: “More accurate information must raise welfare.”

Check. The handwritten derivation will verify the FOC, $g'(Y) < 0$, and the direct/indirect welfare decomposition:

$$\frac{\partial \bar{U}^+}{\partial \tau_A} = DE + IE,$$

$$DE = g(\bar{Y}_h)G(\bar{X}_h)\Delta_X \geq 0,$$

$$IE = \frac{\partial G(\bar{X}_h)}{\partial \tau_A} [\Delta_G + G(\bar{Y}_h)\Delta_X] \leq 0.$$

Verdict: false dynamically. Under Assumption 2, welfare rises only up to τ_A^* , then falls (Props. 10–11, p. 27).

Handwritten photo pending
hand/observation-1-foc.jpg